CEP Mapping

Table 1: How Ks are addressed through the project

Ks	Attributes	How Ks are addressed through the project	Evidence
K3	Engineering fundamentals	The project uses Python programming to build a full RAG pipeline including text extraction, chunking, embedding, and retrieval.	Python code scripts for PDF extraction, text cleaning, IQR filtering, and chunking notebooks.
K4	Specialist knowledge	The project employs advanced NLP techniques like dense vector embeddings (e5-large-v2) and sequence-to-sequence generation (Flan-T5) for university Q&A.	Implementation of hybrid retrieval scoring, FAISS index, and Flan-T5 answer generation pipeline.
K5	Engineering design	An end-to-end RAG pipeline was designed covering PDF ingestion, cleaning, chunking, embedding, retrieval, generation, evaluation, and deployment.	System methodology diagram, chunking strategy, hybrid scoring formula, and structured Colab notebook.
K6	Engineering practice	Practical tools such as HuggingFace Transformers, FAISS, LangChain, Flask, and Google Colab are used for model implementation and deployment.	Colab project notebooks, Python library imports, Flask API code, and ngrok deployment logs.

K7	Comprehension	Use of real university PDF documents following ethical AI principles, with fallback handling to avoid hallucinated answers.	Fallback response ("I don't know based on provided documents"), prompt engineering with context-only instruction.
K8	Research literature	Study of existing research on RAG, dense retrieval, and transformer-based generation that guided architecture and model selection.	Literature-informed choice of e5-large-v2 (Wang et al., 2022), Flan-T5 (Raffel et al., 2020), and FAISS (Johnson et al., 2021).

Table 2 : Complex engineering problem solving (P)

Ps	Attributes	How Ps are addressed through the project	Evidence
P1	Depth of knowledge required	The project implemented advanced NLP (K3), specialist retrieval and generation models (K4), a full RAG pipeline (K5), and literature review on RAG systems (K8).	Technical documentation, training notebooks, hybrid retrieval code, and evaluation results.
P2	Range of conflicting requirements	There are trade-offs between retrieval speed (FAISS) and accuracy (cosine re-ranking), and between answer fluency and factual grounding, requiring careful balancing.	Hybrid scoring formula (0.7 × cosine + 0.3 × FAISS), context length limit (250 words), and fallback handling.

P3	Depth of analysis required	No standard university chatbot with hybrid retrieval existed; abstract thinking was required to design the hybrid scoring system and unified evaluation framework from scratch.	Hybrid retrieval function, failure case analysis, BLEU/ROUGE/semantic similarity evaluation pipeline.
P4	Familiarity of issues	The project works with university-specific document retrieval, an infrequently studied domain with no standard RAG benchmark for institutional Q&A.	Custom JSON Q&A pairs for evaluation, domain-specific PDF knowledge base, IQR-based filtering design.
P5	Extent of applicable codes	The project addresses issues outside existing standards, requiring self-defined chunking strategy, hybrid scoring formula, and custom evaluation rubric.	Custom chunk size (180 words, 40 overlap), self-defined quality tiers (Excellent/Good/Partial/ Poor), custom output cleaning function.
P7	Interdependence	High-level problem with many interconnected sub-problems: PDF ingestion, cleaning, chunking, embedding, indexing, retrieval, generation, evaluation, and deployment.	Full end-to-end pipeline across 46 notebook cells covering all sub-systems as one integrated solution.

Table 3 : Complex engineering activities (A)

As	Attributes	How As are addressed through the project	Evidence
A1	Range of resources	Engaged the use of diverse resources including university PDFs, JSON Q&A datasets, Python libraries (HuggingFace, FAISS, LangChain), and GPU compute on Google Colab.	Google Colab notebooks, HuggingFace model hub, FAISS vector index, LangChain text splitter, Flask REST API.
A2	Level of interaction	Resolution of significant problems arising from retrieval accuracy vs. speed conflict, context length limitation of Flan-T5, and output noise requiring post-processing.	Hybrid scoring (FAISS + cosine), context truncation to 250 words, clean_output() function for prediction normalization.
A3	Innovation	A degree of innovation is needed to design hybrid retrieval scoring combining FAISS inner product and cosine similarity, and to integrate JSON Q&A data into the PDF knowledge base.	Hybrid score formula (0.7 × cosine + 0.3 × FAISS), merged PDF + JSON knowledge base, structured prompt engineering.

A4	Consequences for society and the environment	Automated university Q&A chatbot improves student access to information 24/7, reduces staff workload, and promotes responsible AI grounded only in verified university documents.	Live Flask API deployment, fallback response for out-of-context queries, ethical prompt design restricting hallucination.
A5	Familiarity	The project extends beyond standard NLP coursework by applying dense retrieval, hybrid scoring, and RAG-based generation using principles-based approaches.	e5-large-v2 embedding pipeline, FAISS IndexFlatIP, hybrid retrieval re-ranking, Flan-T5 sequence-to-sequence generation.